BAMBINOS.LIVE · AI PROGRAM MANAGER (OPERATIONS) · TAKE-HOME CASE

Renewals that die in the paperwork

A leakage audit of the renewal to scheduling pipeline, a zero-touch workflow for clean renewals, a working reconciliation engine, and a rollout plan that does not put a live class at risk.

> Twelve parents paid. One child has a class on the calendar. The gap is not effort. It is a form that duplicates a payment record, and a step after approval that nobody owns.

Prepared November 2026 · data as of 3 November 2026, 18:00 · 13 renewal cases, 12 with money received · 32 tests, 6 charts, 1 dashboard

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A · EXECUTIVE SUMMARY

The pipeline leaks in three places, and none of them is effort

16.7% paid invoices with no teacher form 2 of 12, INR 29,000	**91.7%** paid renewals with no future class 11 of 12, INR 126,850, proxy	**27.1 min** median wait for a form n=10, worst 49.8
75.0% paid renewals that need no human 9 of 12, zero-touch eligible	**144 h** ops hours saved per month at 1,000 renewals, base case	**4.3 mo** payback on the build INR 2,00,000 one time

The pipeline is not slow because Ops is slow. In all four cases where a form arrived and Ops approved it, the approval and the booking request were created in the same minute. The delay sits upstream in a form that repeats information the payment already holds, and downstream in an allocation step that has no named owner.

On this data, 2 of 12 paid invoices never got a form at all. Another 8 of 12 forms were still unapproved at the snapshot, the oldest for 32.5 hours. Two of the four bookings never produced a session. The end state is 11 paying students with no future class in the schedule export, worth INR 126,850 of the INR 138,850 collected.

The fix is to stop treating the teacher form as the trigger. For a same-batch renewal the invoice already carries the student, the class, the batch, the teacher, the amount and the discount. The only field the form adds is a preferred time, and in 11 of 12 forms that time is identical to the batch the child is already in. The form is not a control, it is a queue.

I built the engine that acts on that: 700 lines of Python, no runtime dependencies, 32 tests covering every reason code, an n8n export and a Google Sheets spec that mirror the logic, and three prompt files. It auto-approves 7 of 13 cases today and would clear 9 of 12 paid renewals with no human. The five blocking exceptions it leaves have an owner, an SLA and a drafted message.

That is worth about 144 ops hours a month at 1,000 renewals, INR 43,186, against a one-time build of about INR 2,00,000. The ask is small: three weeks in shadow mode, and one live-roster validation that confirms the same-batch renewal retains an existing seat.

A · METHOD

What is in the files, and what they cannot tell me

Five files, 44 rows. I read every row before writing a rule, because a rule built on an assumed schema is how you get a matcher that scores the wrong invoice at 0.97.

FILE	ROWS	KEY	WHAT IT IS
1_invoices.csv	13	id	Money received, gateway reference, payer contact
2_faculty_form_unapproved.csv	12	id	The teacher's claim about the same payment
3_booking_requests.csv	4	id	Ops approval, one row per approved form
4_class_schedule_bookings.csv	2	id	Actual sessions, the only proof a class exists
reference_active_batches_and_students.csv	13	student_id	Active batch, teacher, fee, capacity

Joins, ranked by how much I trust them. **student_id** is exact in 13 of 13 invoices and 12 of 12 forms. **invoice_id** is exact in 10 of 12 forms; one is a typo (84931 against a real 84913) and one is a duplicate submission. **faculty_form_id** links only 4 of 12 forms to a booking. **bookingrequest_id** reaches only one of four bookings. The invoice link is well formed in all 12 forms but must never be used as a key, because two invoice URLs differ by three characters out of fifty.

form_submission_lag is measurable for 10 forms, **approval_lag** for 4, and **payment to schedule** for exactly 1. I report the sample size next to every timing number, because a median of one is an anecdote.

> **What this data cannot settle.** All names, phones and emails are masked, so fuzzy name matching cannot be validated here. There are no term start or end dates, so class interruption is a proxy: money received and no future session row. There is no gateway settlement feed, so a late payment or a chargeback is invisible. And the schedule table has 2 rows for 12 paid renewals, which is either a catastrophe or a partial export. I treat it as partial and say so wherever the number appears.

Three days of data, 13 cases, one Sunday to Tuesday window. Every monthly figure in this report is a scenario with the volume assumption written next to it, not a measurement.

A · LEAKAGE AUDIT

Where students get stuck, stage by stage

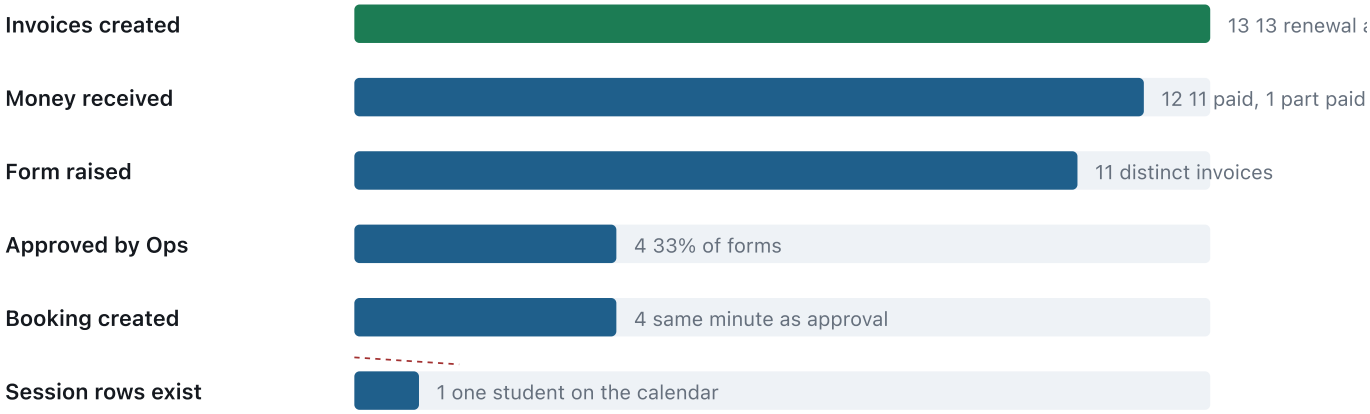


Figure 1. The renewal funnel on 13 cases. Counts are exact; the final bar is a partial schedule export.

Thirteen invoices were raised. Twelve collected money. Eleven distinct invoices got a form. Four were approved. Four bookings were created. One student has session rows.

Each drop has a different cause, which matters because no single fix would have caught all of them. The two orphans needed no form at all: one is a GCC parent who paid INR 18,000, the other a chess parent who paid INR 11,000, and neither teacher submitted anything. The eight unapproved forms needed a decision that never came. The three unscheduled bookings exposed two allocation failures and one false batch-full rule.

The shape of the funnel is the argument for the redesign. The middle of the pipeline, where a human retypes a payment into a form and another human checks it, holds no information the system did not already have. It only adds waiting, and two ways to lose a child: the form never arrives, or the approval never happens.

A · FAILURE MODES

Fourteen hypotheses, counted, with the money attached

Every hypothesis in the brief, tested, with the denominator stated. The rows that came back zero stay in the table because a rule that fires on clean data is worse than no rule.

FAILURE MODE	COUNT	DENOMINATOR	VALUE AT RISK	EVIDENCE
Paid invoice, no teacher form	2	12 paid, 16.7%	INR 29,000	84906, 84910
Approved, never scheduled	2	4 bookings, 50%	INR 21,500	5502 idle 51.7h, 5505 failed
Form names a missing invoice	1	12 forms, 8.3%	INR 11,500	3313 points at 84931
Invoice used in two forms	1	11 invoices, 9.1%	double booking risk	3308 and 3309, 5.2 min apart
Claimed amount differs	1	12 forms, 8.3%	INR 5,250	3310 claims 10,500 against 5,250
Part payment	1	13 invoices, 7.7%	INR 5,250	84909, installment 1 of 2
Failed payment	1	13 invoices, 7.7%	INR 12,000	84911, no gateway payment id
Payer contact mismatch	1	12 paid, 8.3%	INR 12,000	84904 paid from XXXX77, record has XXXX04
Batch at capacity	1	12 forms, 8.3%	INR 11,000	9205 at 6 of 6
Batch change	1	12 forms, 8.3%	INR 9,000	3312 wants Sat/Sun 11:00
Student missing from reference	0	13 students	0	all present
Malformed invoice URL	0	12 forms	0	all 12 well formed
Duplicate schedule rows	0	2 rows	0	distinct dates
Currency mismatch	0	12 paid	unknown	no currency field in the reference table

Five things the brief did not ask for

The batch-full block is a counting artefact. Batch 9205 is at 6 of 6 and student 68105 is one of those six. A same-batch renewal extends that occupied seat rather than taking a seventh, so the engine retains it automatically while keeping a hard capacity block for a genuinely new allocation.

Approval is not the bottleneck. All four approved forms produced a booking in the same minute. Then booking 5502 sat at Pending_Batch_Allocation for 51.7 hours. A faster approval inbox would move the wrong number.

A discount looks like an underpayment. Form 3307 claims 7,600 against a 9,500 list price. The invoice records a 1,900 discount, exactly 20%, so the claim matches the net amount. Any rule that compares the form to the gross price flags every scholarship renewal.

The form is not independent evidence. On 84904 the invoice carries a number that appears nowhere else, and the form supplies the one that matches the parent record. The teacher typed it from the same WhatsApp thread that produced the payment, so it is a copy of the conversation, not a second source.

Cross-border money is unreconciled. Two Stripe payments, from a UAE and a US number, both billed in INR for INR 40,000 combined. There is no currency field in the reference table, so nothing in this data can confirm the settlement amount.

B · PROCESS AUDIT

Why each manual step exists, and which one to delete

The brief asks whether the teacher form can be eliminated for clean renewals. The answer is arithmetic, not opinion. Take the four fields the form exists to capture and ask where else they live.

STEP	STATED PURPOSE	DOES IT HOLD	DECISION
Invoice created by the gateway	Record that the parent paid	Holds. It is the only source of truth for money.	Keep, and promote to the trigger for the whole flow.
Teacher fills a renewal form	Capture student, batch, amount, preferred time	Four of five fields are already on the invoice. The fifth matches the current batch in 11 of 12 forms.	Remove as a trigger. Keep as corroboration and as the carrier of a batch-change request.
Ops approves the form	Verify the payment before booking	Verification is three field comparisons. Approval and booking creation landed in the same minute in all four cases.	Replace with rules. Humans keep capacity, identity and policy decisions.
Batch allocated, sessions created	Parent or Ops picks days and time	For a same-batch renewal there is nothing to pick. Booking 5502 waited 51.7 hours for a click.	Auto-allocate same-batch. Escalate only batch changes.

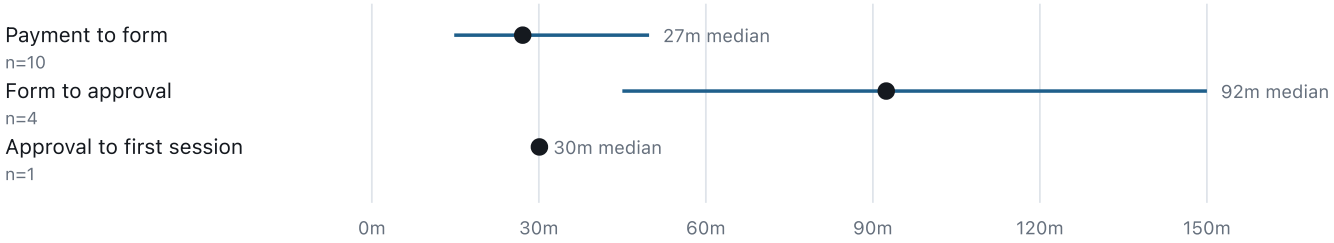


Figure 2. Where the 105 minutes went on the one case that completed. Approval was 43% of it.

The only field the form genuinely adds is the preferred day and time. In 11 of 12 forms it is identical to the batch the student is already in, because the parent is renewing the same slot with the same teacher. The twelfth is a real batch change, and that is the one case where a form is still worth collecting.

So the redesign does not delete the form. It demotes it. The invoice triggers the flow, the form becomes a piece of evidence that can block an edge case but never gates a clean one, and the queue that produced 27 minutes of median waiting and a 16.7% orphan rate stops existing.

B · RE-ENGINEERED WORKFLOW

One trigger, two lanes, no form in the critical path

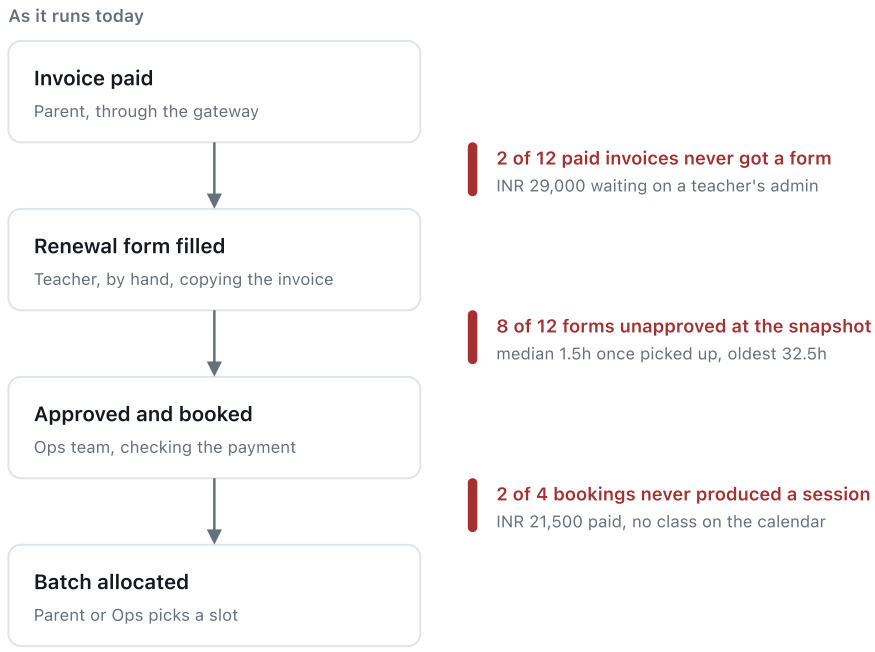


Figure 3. Today. Three handoffs, three different ways to lose a paying child.

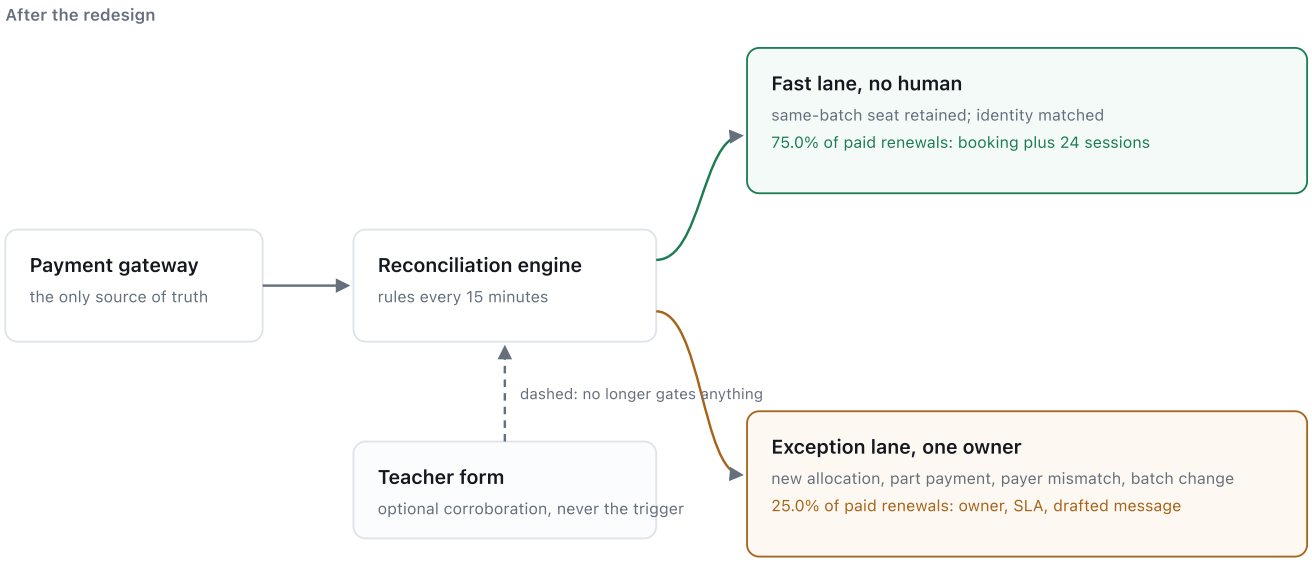


Figure 4. After. One trigger, two lanes, and a form that no longer gates anything.

The fast lane is deliberately narrow. All four conditions must hold: the invoice is Paid or Partially_Paid with the net equal to billed minus discount; the student, at least one contact channel, the teacher and the class match the active record; and the renewal is same-batch and the form, when present, names that batch. That continuation retains its existing seat. Any new allocation must reserve capacity; if it cannot, the case goes to the exception lane. It never half-approves.

In the exception lane the case keeps everything the fast lane would have given it: the resolved invoice, the batch, the amount, the payer, and a sentence explaining exactly which check failed. What changes is that a named person now owns it, with a clock running.

B · EXCEPTION HANDLING

Every failure has one owner and a clock

Each exception type goes to the person who can actually change its state. A teacher cannot see capacity, so a teacher never owns a capacity decision. Finance cannot call a parent about a class, so finance owns the money question and Ops owns the message.

SITUATION	CODE	OWNER	REPLY / RESOLVE	FIRST ACTION
Batch at capacity	R08	ops_agent	4h / 8h	Ask the teacher whether the batch continues. If yes, renew in place. If no, offer a batch with a free seat.
Part payment	R06	finance	4h / 24h	Agree a balance date or pro-rate the sessions. Do not release 24 sessions on half the fee.
Payer contact mismatch	R12	ops_agent	4h / 4h	One call to confirm the payer, then release. The parent also gets a message so the wait is explained.
Batch change	R09	parent	8h / 24h	Confirm days, time and teacher, then allocate. The only case where the form is still the trigger.
Payment not completed	R07	parent	24h / 48h	Send the retry link, hold the class, trace the reference if money left the account.

The escalation ladder is a clock, not a judgement call

LEVEL	ROLE	TRIGGER	AUTHORITY
1	Assigned owner	Case enters the queue	Resolve within policy
2	ops_lead	SLA breached or value above INR 15,000	Reassign, approve a seat override with teacher consent
3	Head of Operations	Breach over 24h, or any child misses a class	Approve refunds, credits, batch creation
4	Finance lead	Payment dispute or chargeback	Owns gateway reconciliation

What the parent actually receives

WHATSAPP TO PARENT · R12 · DRAFT FROM OUTPUTS/MESSAGES.CSV

Hi, this is Priya from Bambinos. An***a Iy***r's renewal of Rs 12,000 came through on 2 Nov. The payment shows +91-9844XXXX77, which is not the number we have on file, so I want to confirm it is yours before I release the classes. Is that the right number? Reply yes and I'll get the classes scheduled today.

Three sentences, one specific number, no blame, and a promise the ops agent can keep. The full draft set is 12 messages across 5 cases, in `outputs/messages.md`.

C · METRICS AND SLAS

Eight numbers, and the one that actually matters

Eight metrics, each with a formula an ops lead can recompute from the same files. Baselines come from this dataset at 2026-11-03 18:00. Where the sample is too small to be a baseline, I say so rather than quoting it as one.

METRIC	BASELINE TODAY	TARGET	OWNER
Orphaned Invoice Rate	16.7% (2 of 12)	under 2%	ops_agent
Payment to Schedule	1.75h, n=1; 11 of 12 open	median under 15 min, p90 under 4h	ops_agent
Form Submission Lag	27.1 min median, n=10	not applicable once the form is optional	teacher
Approval Lag	1.54h median, n=4; 8 of 12 unapproved	under 5 min auto, 4h exception	ops_agent
Unscheduled Booking Rate	75% (3 of 4)	under 5% at 24h	ops_agent
First Pass Auto-Approval Rate	0%	70%	Head of Ops
Exception Resolution Time	8.0h median open age, n=5	median 4h, p90 24h	assigned owner
Class Interruption Rate	91.7% (11 of 12), proxy	under 2%	Head of Ops

The one that matters is the last. Every other metric is a leading indicator; class interruption is the thing the parent experiences. It is also the number I trust least, because the schedule export has two rows. If a full export shows the same rate, this stops being a process problem and becomes an emergency.

> **Deliberately not an SLA.** Teacher response time, because teachers are paid per class and policing their admin would need a contract change. Parent response time, because a parent who paid and stopped replying is not a process failure. Time to first class, because it depends on the batch calendar, not on Ops.

Escalation is automatic at the SLA. The engine stamps `sla_hours` on every row, the dashboard counts down against it, and the daily digest names whatever is past due. Nobody has to decide whether a case is late.

C · PROTOTYPE

Deterministic rules, with the LLM kept on a short leash

Five CSVs in, six operational artifacts out, one command. The pipeline is load, normalize, match, decide, allocate seats, write. No runtime dependencies, no network calls, no wall clock in the outputs.

The rule order, which is also the blast radius

1. Resolve the invoice: exact id, then link, then fuzzy repair with a uniqueness requirement.
2. Payment settled: status must be Paid or Partially_Paid.
3. Fee settled: net must equal billed minus discount, within tolerance.
4. Claim matches: the form amount must equal the net collected.
5. Identity: student id, then at least one contact channel.
6. Batch: same-batch, active, correct teacher and class, valid session count.
7. Seats: same-batch renewals retain an existing seat; new allocations reserve one through a ledger.
8. State: already scheduled, approved but unscheduled, or approved with no booking.

PARAMETER	VALUE	WHY THAT VALUE
Invoice similarity floor	0.72	84931 to 84913 scores 0.80; unrelated numbers score near 0
Runner-up margin	0.05	refuse when two candidates are close, rather than guessing
Amount tolerance	INR 1 or 0.5%	gateway rounding, nothing more
Stuck invoice threshold	12 hours	the dashboard tile the ops head asked for
Escalation clock	24 hours	one business day

What it refuses to do. It will not approve against a failed payment, even with a screenshot. It will not pick between two plausible invoices; it returns R02 and asks a teacher for the number. It will not edit an invoice amount to match a form. And it will not put a second booking on an invoice that already has one.

Every row carries a reason code and a sentence with the numbers in it. That is the difference between a script and something an ops team can trust at 9am on a Monday.

C · PROTOTYPE RESULTS

Eight rows cleared, five blocking exceptions, one refusal I would defend

Seven cases auto-approved, five blocked, one duplicate closed, one already scheduled. The seven rows in `auto_approved.csv` are ready to become bookings: five came from a form, two from an invoice with no form at all, and one from a repaired reference.

CASE	EVIDENCE	BATCH	SEATS LEFT	FLAGS
84903	form + invoice	9203	3	-
84905	form + invoice	9205	0	-
84906	invoice only	9206	1	R01,R18
84907	form + invoice	9207	2	R22
84908	form + invoice	9208	1	R18
84910	invoice only	9210	1	R01
84913	form + invoice	9213	2	R03

CODE	EXCEPTION	COUNT	OWNER
R04	Duplicate form for the same invoice	1	teacher
R06	Part payment received	1	finance
R07	Payment not completed	1	parent
R09	Batch change requested	1	parent
R12	Payer contact does not match the student record	1	ops_agent
R20	Approved but never scheduled	1	ops_agent

The decision I would defend hardest is a refusal. Case 84904 was paid from +91-9844XXXX77, which matches neither the registered mobile (+91-9844XXXX04) nor the registered email. The teacher's form supplies the 04 number, and the remark says the mother paid. I still held it, for one call. Auto-approving would have saved two minutes and risked a misapplied payment on a child whose parent would then have to argue about it.

Case 84913 shows the opposite. The form typed invoice 84931 and the real invoice is 84913. The matcher repaired it because the student, the amount and the teacher all matched and there was exactly one candidate. That is a class cleared in milliseconds that a human would have bounced back to the teacher.

Two of the seven auto approvals exist only because the form was demoted. Their teachers never submitted anything, and the payment alone was enough: INR 29,000 that the current process would still be chasing.

C · DRAFTED MESSAGES

Short, specific, and addressed to one person

Twelve drafts across five cases, each addressed to one person, each containing the number that triggered it. These are the exact strings in `outputs/messages.csv`, produced by templates with no API key.

PAYER MISMATCH, TO THE PARENT · WHATSAPP TO PARENT

Hi, this is Priya from Bambinos. An***a Iy***r's renewal of Rs 12,000 came through on 2 Nov. The payment shows +91-9844XXXX77, which is not the number we have on file, so I want to confirm it is yours before I release the classes. Is that the right number? Reply yes and I'll get the classes scheduled today.

PART PAYMENT, TO FINANCE · SLACK TO FINANCE

84909 Ad***t Ku***i. Billed Rs 10,500, net_amount Rs 5,250, payment_type Installment_1_of_2. The teacher form claims Rs 10,500 paid, which is the full fee, not what landed. Do not release 24 sessions on this. Decide pro-rated sessions or hold for the balance. Open 8.0 hours.

BATCH CHANGE, TO THE PARENT · WHATSAPP TO PARENT

Hi, this is Priya from Bambinos. Ta***a Me***n's Rs 9,000 renewal is through, thank you. You asked for Sat and Sun at 11:00 instead of Wed and Fri at 16:30, and I have that noted. I'm matching the student to a batch now. Two things: can you confirm 11:00 works for the whole term, and should the teacher stay Acharya Nancy S***? I'll confirm the batch by tomorrow evening.

FAILED PAYMENT, TO THE PARENT · WHATSAPP TO PARENT

Hi, this is Priya from Bambinos. Is***n Gu***a's renewal payment of Rs 12,000 did not go through on 3 Nov. Nothing has been charged. The classes are on hold until it clears, so please retry the same link when you get a minute: https://bambinos.live/invoice/BMB-INV-2026-84911. If money did leave your account, send me the reference and I'll trace it today.

Where the LLM sits. Three prompts in `prompts/`. One polishes message copy, one resolves genuinely ambiguous invoice references, one triages an exception that the rule set has not seen before. All three take a facts payload and are forbidden from adding a number, a date or a promise. The default run does not call any of them, and the output is identical without a key.

That is a deliberate cost decision as much as a safety one. Message drafting is the only place where wording quality matters and a wrong answer is cheap. Decisions are not.

D · OPS DASHBOARD

What the Head of Operations sees at 9am

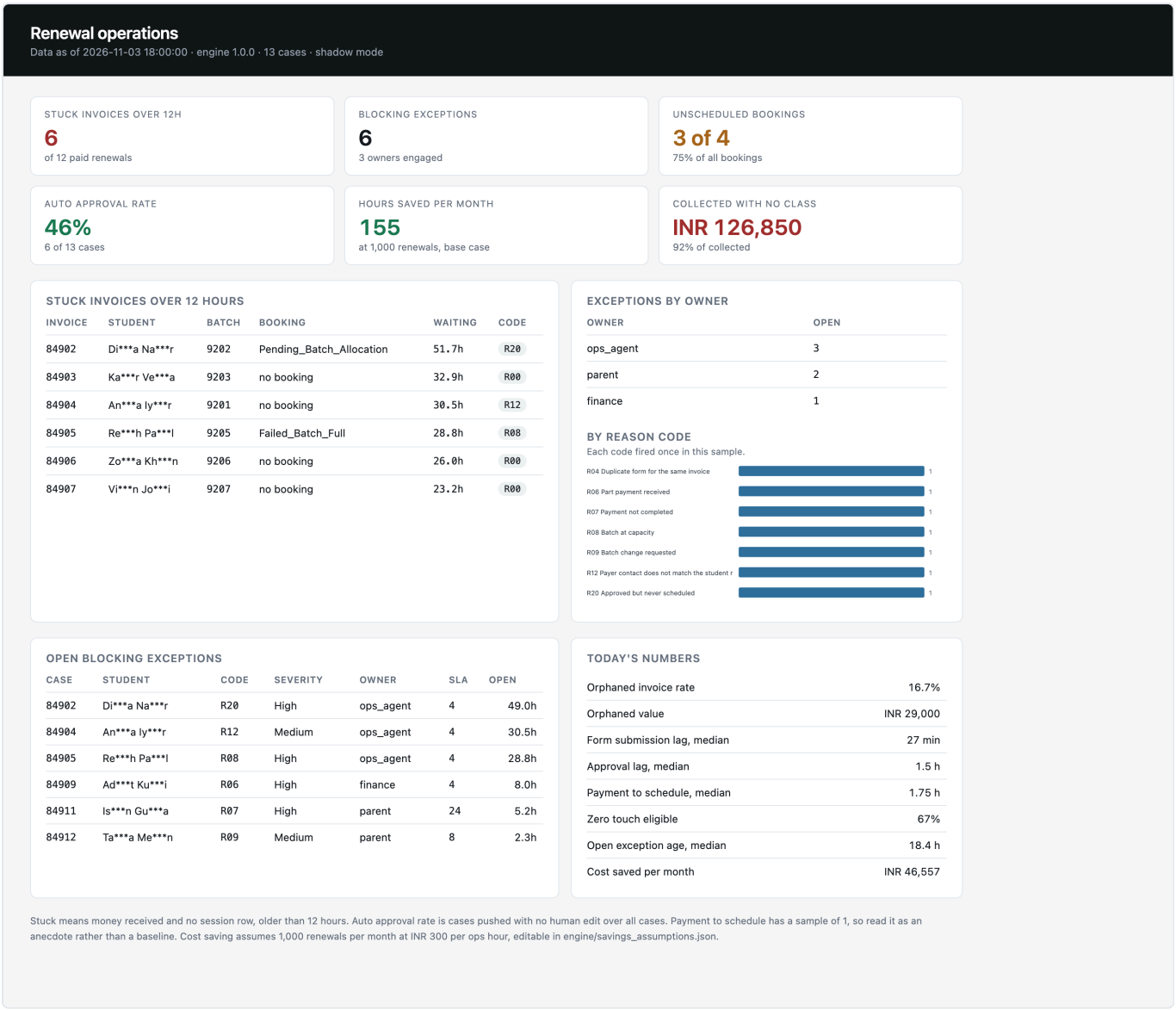


Figure 5. The dashboard, built from the engine outputs, not mocked up.

TILE	DEFINITION	TODAY
Stuck invoices over 12h	Money received, no session row, older than 12 hours	6
Blocking exceptions	Rows where a human must change state before a class exists	5
Unscheduled bookings	Bookings with zero session rows	3 of 4
Auto approval rate	Cases pushed with no human edit, over all cases	54%
Hours saved	Monthly ops hours avoided at the modelled volume	144
Collected with no class	Net revenue on invoices with no future session	INR 126,850

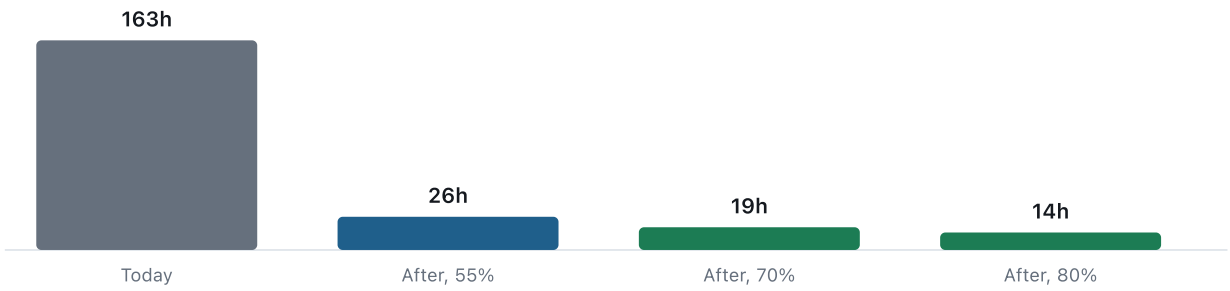
The morning routine is twenty minutes, not a dashboard tour. Read the six tiles, then clear your own exceptions oldest first, because the oldest is a parent who has waited longest. Then spot-check five auto-approved rows, flagged ones first. If one is wrong, stop the engine before fixing it by hand.

The tile I would watch is not the exception count. It is the age of the oldest open exception. A stable count with a rising age means the queue is being triaged but not cleared, which is exactly how the current process fails: 8 forms pending, none of them moving.

D · SAVINGS MODEL

144 hours a month, with every assumption visible

Ops hours per month at 1,000 renewals



Base case saves 144 hours a month, about INR 43,186 at INR 300 an hour.

Figure 6. Ops hours per month at 1,000 renewals. Today is 162.7 hours; the base case is 18.7.

ASSUMPTION	VALUE	WHY
Monthly renewal volume	1,000	Scenario. The file shows 13 cases over 3 days
Ops loaded hourly cost	INR 300	About INR 54,000 a month at 174 productive hours
Context switching allowance	25%	Task switching and re-reading threads, added to manual time
Automation overhead	10%	Digest review and the 5% audit sample
Auto approval rate	70%	Observed 75.0% on this data, modelled lower for sample noise
One time build	INR 2,00,000	Six weeks of one engineer plus part of an ops lead

AUTO APPROVAL RATE	HOURS SAVED / MONTH	COST SAVED / MONTH	PAYBACK
55%	137	INR 41,067	4.9 mo
70%	144	INR 43,186	4.6 mo
80%	149	INR 44,598	4.5 mo

Today the same 1,000 renewals cost 162.7 ops hours: 7.81 minutes of task time per case, inflated by a 25% context-switch allowance. The redesign lands at 18.7 hours, because 70% of cases need a quarter of a minute of oversight and the remaining 30% need 3.1 minutes each.

What is not in the number. Recovered revenue. If faster scheduling stopped even 2% of at-risk renewals from lapsing, that would add roughly INR 2.8 lakh a month at this average net, which dwarfs the labour saving. I left it out because the churn rate is invented and the labour saving is not.

> Every input sits in `engine/savings_assumptions.json`. Change the volume to 300 and the hours scale linearly; the payback moves to 14 months. Nothing in the model is hidden.

D · ROLLOUT

SOPs, 30-60-90, and how a rollback cannot break a class

Teachers. The SOP is five things they may be asked, four rules that save them time, and a list of what we will never ask them to do, including chase a parent for money. The headline for them is that the form is gone for same-batch renewals, so this rollout removes admin rather than adding it. That is the whole adoption argument.

Ops. The SOP replaces the approval inbox with an exception queue. Work oldest first, close on state change rather than on reply, and never edit an invoice amount to match a form. The weekly review is 30 minutes on cases that took more than 24 hours, because three weeks of that list tells you which rule to change next.

WINDOW	FOCUS	GATE TO PASS
Days 0 to 30	Shadow mode. Engine recommends, Ops decides. Every disagreement logged with one line of why.	Zero disagreements on the last 50 cases of a code before it can go live.
Days 31 to 60	Pilot with two teachers and about 20% of volume. Only R00 and R17 write.	No wrong child scheduled, auto-approval rate within 10 points of prediction.
Days 61 to 90	Phased cutover by reason code. Ops keeps the exception lane.	Median exception age under 4 hours for two consecutive weeks.

Keeping live schedules safe

The engine never edits an existing session row. It writes bookings through one endpoint, and only for invoices with no existing booking. Before any write it produces the same rows in shadow mode so the diff can be read. Rollback is one switch that returns writes to manual while the engine keeps recommending, so a rollback cannot orphan a class that already exists.

Rollback criteria, written before launch. More than 0.5% false approvals in the audit sample. Any case where the wrong child is scheduled. More than two escalations a day caused by the engine rather than by the parent. Any breach of the 12 hour stuck threshold that the engine created rather than inherited.

The pilot is two teachers, not two hundred, because the failure modes I care about are social: a teacher who keeps filling forms out of habit, an ops agent who overrides without a reason. Both show up in week one with two teachers and are invisible at scale.

D · RISKS AND LIMITS

What breaks, what I would do differently, what I am not claiming

RISK	LIKELIHOOD	IMPACT	MITIGATION
Wrong child scheduled	Low	High	Hold on dual contact mismatch, 5% audit sample, refusal by default when two invoices fit
False batch-full block	Low	Medium	Retain the active student's seat; require a capacity reservation only for a new allocation
Teacher resistance	Medium	Medium	The form is removed, not added. No new admin for teachers
Schedule export incomplete	High	High	Verify in week one. Do not publish the interruption rate until it is real
Volume breaks the Sheets version	Medium	Medium	Move to Postgres and a queue. The rules are stateless and linear
Overrides drift	Medium	Medium	Override reason required, reviewed weekly, counted on the dashboard

What I would do differently

I would start from the schedule table, not the invoices. The funnel looks like an approval problem until you notice that approval and booking creation happen in the same minute, and the real gap is allocation. Reading the last table first would have found that in ten minutes instead of an hour.

I would also instrument time-in-state from day one. Every number in this report is reconstructed from timestamps that happen to exist, and two of the eight metrics have a sample size small enough that I have to apologise for them. A single `state_changed_at` column would make all eight measurable.

And I would ask for one unmasked sample of 50 rows. The masked data is honest about structure and useless for validating the one thing that needs it, which is whether two names that look alike belong to the same child.

What I am not claiming

This is a 13 case sample, not a production baseline. The interruption rate rests on a two row schedule table. The LLM path has never run against a real key. The n8n export is structurally valid and its JavaScript passes a syntax check, but it has not been imported into a live instance. The Sheets spec has not been pasted into a sheet.

What I am claiming is narrower and, I think, more useful: on the data provided, three quarters of paid renewals need no human, the five that do have a name and a clock, and every decision the machine makes can be read and argued with by the person who owns the class.